OFFICIAL SPECIFICATION & TASK CONFIRMATION • AIT-201 APPLIED ML

PERSON 3: DATA ARCHITECT & PIPELINE LEAD

End-to-End Technical Execution Plan: Single-Source Batch Pipeline (BASE) & Multi-Source Canonical Simulation (UPGRADE)

ROLE / ASSIGNEE	**Person 3 • Data Architect & Pipeline Lead**	STATUS	**CONFIRMED • READY TO EXECUTE**
PHASE 1 (BASE)	Goal 4 Base (Single-Store ETL → lots_base.csv)	BOTTLENECK	**NONE** (100% Day 1 Independent Work)
PHASE 2 (UPGRADE)	Goal 3 (Canonical Schema) + Goal 4 Full (Event Simulator)	LOCALIZATION	Malaysia (MYR Currency, Klang Valley, UTC+8)
CORE ARTIFACTS	process_store_inventory.py, validate_lots_base.py, lots_base.csv	ANCHOR DATE	2026-10-30 (Aligned with Frontend & DOSM)

1. Role Mandate & Strategic Architectural Positioning

1.1 Definitive Mission: The Foundation & Single Source of Truth

Dalam arsitektur *Dynamic Food Clearing House*, Person 3 (Data Architect & Pipeline Lead) memegang peran vital sebagai **penyedia data tunggal yang berwibawa (Single Source of Truth)**. Seluruh model machine learning dan logika aplikasi — Model Klasifikasi Kerusakan FEFO (Person 1), Peramalan Permintaan (Person 2), Rule Engine Diskon Dinamis & Cloud DB Supabase (Person 4), Scoring Risiko Waste & Smart Dispatch OSRM (Person 5), serta Web Dashboard Toko 6-Langkah (Person 6) — **100% bergantung langsung pada kebersihan, format standar, dan konsistensi skema tabel yang dihasilkan oleh pipeline Person 3**.

PANDUAN STRATEGIS & RESOLUSI FAQ RESMI:
• **Apakah di Fase 1 (BASE) Person 3 harus menggabungkan 5 dataset sekaligus?**
TIDAK SAMA SEKALI! Sesuai dokumen FAQ.md dan PEMBAGIAN_TUGAS.md, di Fase 1 fokus Anda **hanya** mengolah 1 dataset inventaris mentah (Grocery_Inventory new v1.csv) menjadi **lots_base.csv**. Pekerjaan menggabungkan 5 dataset (Canonical Schema) adalah tugas Fase 2 (UPGRADE).
• **Mandat Eksekusi Bebas Bottleneck (Day 1):** Anda dapat langsung memproses data inventaris lokal tanpa menunggu anggota lain.

1.2 Diagram Aliran Data & Interkoneksi Modul Tim:

```
[Input Data: Grocery_Inventory new v1.csv / cleaned] → [src/etl/process_store_inventory.py]
■■■ 1. Fix Typo 'Catagory' → 'Category' & Cabbage NaN → 'Produce'
■■■ 2. Currency Cleansing: Regex '$' → Float → unit_price_myr (MYR)
■■■ 3. Date Harmonization: 'M/D/YYYY' → ISO-8601 'YYYY-MM-DD'
■■■ 4. Temporal Anchoring: Kalibrasi DTE (1-12 hari) relatif thd Anchor Date (2026-10-30)
■■■ 5. Feature Engineering: shelf_life_days, turnover_ratio, stock_pressure
↓
[OUTPUT BASE UTAMA: data/processed/lots_base.csv]
■■■→ Person 1 (ML): spoilage_classifier.pkl & sort_by_fefo()
■■■→ Person 2 (API): FastAPI GET /api/lots & demand matching
■■■→ Person 4 (BE): Supabase 'batches' Table Ingestion & assign_route()
■■■→ Person 5 (GIS): OSRM Dispatch Route, polyline & food bank matching
■■■→ Person 6 (FE): 6-Step Store Dashboard & Leaflet Route Preview
```

2. Phase 1 (BASE) Technical Specification — Single-Store Batch Pipeline (Goal 4 Base)

2.1 Aturan Pembersihan Data & Rekayasa Fitur (Cleansing Rules):

- **Typo & Null Handling:** Standarisasi kolom header Catagory → Category. Imputasi baris 574 Cabbage (null) secara terarah dengan 'Produce' berbasis pengetahuan domain retail.
- **Pembersihan & Konversi Mata Uang:** Kolom mentah Unit_Price berformat string dengan simbol dollar ($4.60). Diekstrak nilai numerik menggunakan Regex [^0-9.] lalu dikonversi ke Ringgit Malaysia (kurs 1 USD ≈ 4.45 MYR) menjadi kolom numerik unit_price_myr.
- **Standardisasi Format Tanggal:** Konversi seluruh format tanggal M/D/YYYY ke ISO-8601 (YYYY-MM-DD) agar seragam dengan Supabase PostgreSQL dan JavaScript frontend.
- **Strategi Temporal Anchoring:** Dataset Kaggle statis bertanggal 2024–2025. Jika dihitung ke tahun 2026, days_to_expiry akan negatif dan merusak alur FEFO. Pipeline menerapkan **Anchor Date** (2026-10-30) dan kalibrasi siklus 1–12 hari expiry sehingga tercipta sebaran realistis untuk mendemonstrasikan diskon dinamis dan donasi pangan.
- **Pre-Computation Fitur ML (Kontrak Person 1):** Menghitung fitur turunan shelf_life_days = (Expiration_Date - Date_Received).days, turnover_ratio = Inventory_Turnover_Rate, dan stock_pressure = Stock_Quantity / (Reorder_Level + 1).

2.2 Kamus Skema Kolom Deliverable Utama: data/processed/lots_base.csv

Field Name	SQL / Py Type	Constraints	Description & Business Meaning	Sample Value	Consumers
`lot_id`	VARCHAR(32)	**PK, NOT NULL**	Identifier unik tiap lot stok barang.	LOT-1001	Person 1, 2, 4, 5, 6
`product_name`	VARCHAR(128)	**NOT NULL**	Nama komoditas bahan pangan, Title Case.	Organic Spinach	Person 2, 4, 6
`category`	VARCHAR(64)	**NOT NULL**	Kategori: Produce, Dairy, Meat, Bakery, Pantry.	Produce	Person 1, 4, 5, 6
`quantity`	INTEGER	**NOT NULL, > 0**	Kuantitas fisik stok dalam unit atau kilogram.	46	Person 1, 2, 4, 5, 6
`expiry_date`	DATE (ISO)	**YYYY-MM-DD**	Tanggal batas kedaluwarsa resmi produsen.	2026-11-02	Person 1, 2, 4, 6
`unit_price_myr`	NUMERIC(8,2)	**NOT NULL, > 0**	Harga jual reguler per unit dalam Ringgit Malaysia.	20.47	Person 4 (Markdown), 6
`store_id`	VARCHAR(32)	**NOT NULL**	Kode cabang supermarket di Lembah Klang.	MY-KL-001	Person 2, 4, 5, 6
`shelf_life_days`	INTEGER	**>= 1**	Total masa simpan (Expiration Date - Date Received).	8	Person 1 (Spoilage ML)
`days_to_expiry`	INTEGER	**>= 0**	Sisa umur lot terhadap tanggal operasional (DTE).	3	Person 1, 4, 6
`turnover_ratio`	NUMERIC(6,2)	**>= 0.0**	Kecepatan perputaran stok barang di gudang ritel.	1.96	Person 1 (Spoilage ML)
`stock_pressure`	NUMERIC(6,2)	**>= 0.0**	Rasio stok saat ini terhadap Reorder Level toko.	0.71	Person 1 (Spoilage ML)

2.3 Cuplikan Logika Script ETL (src/etl/process_store_inventory.py):

```
def process_store_inventory(input_path, output_path, anchor_date=datetime(2026, 10, 30)):
df = pd.read_csv(input_path).rename(columns={'Catagory': 'Category'})
df.loc[(df['Product_Name'] == 'Cabbage') & (df['Category'].isna()), 'Category'] = 'Produce'
df['unit_price_myr'] = (df['Unit_Price'].replace(r'[\$,]', '', regex=True).astype(float) * 4.45).round(2)
df['shelf_life_days'] = (pd.to_datetime(df['Expiration_Date']) - pd.to_datetime(df['Date_Received'])).dt.days.clip(lower=1)
df['days_to_expiry'] = ((df.index % 12) + 1) # Calibrated cycle 1-12 DTE relative to anchor
df['stock_pressure'] = (df['Stock_Quantity'] / (df['Reorder_Level'] + 1)).round(2)
df.to_csv(output_path, index=False)
```

3. Data Quality Assurance & Automated Validation Suite (`validate_lots_base.py`)

Untuk menjamin tidak ada data anomali yang lolos ke database Supabase atau pipeline ML, Person 3 membangun skrip otomatis src/etl/validate_lots_base.py yang memverifikasi **7 Golden Rules of Pipeline Integrity**:

No	Integrity Rule	Verification Logic & Condition	Defensive Action on Fail
1	**Zero Missing Primary Keys**	`df['lot_id'].isna().sum() == 0`	Tolak batch, raise ValueError.
2	**Primary Key Uniqueness**	`df['lot_id'].is_unique == True (zero duplicates)`	Hentikan pipeline, cegah duplikasi Supabase.
3	**Strict Positive Quantities**	`(df['quantity'] > 0).all()`	Filter stok nol/negatif ke error log.
4	**Sensible Price Boundary**	`(df['unit_price_myr'] >= 0.50).all()`	Imputasi median kategori DOSM PriceCatcher.
5	**ISO-8601 Date Parsing**	`Regex ^\d{4}-\d{2}-\d{2}$ match on all rows`	Konversi paksa dengan parser ISO fallback.
6	**Non-Negative DTE**	`(df['days_to_expiry'] >= 0).all()`	Clip nilai negatif ke 0 (expired today).
7	**Category Taxonomy Match**	`df['category'].isin(APPROVED_CATEGORIES)`	Mapping ke 'General Grocery'.

4. Phase 2 (UPGRADE): Canonical Schema & Discrete-Event Simulator Engine

4.1 Goal 3 — Canonical Schema 5 Dataset (`canonical_batch_table.csv`):
Pada Fase Upgrade, sistem berkembang dari 1 toko menjadi simulator clearing house multi-cabang. Person 3 merancang skema kanonikal yang mengharmonisasikan 5 dataset heterogen lintas domain:

Canonical Cat	Grocery Inventory	Perishable Goods	M5 / Favorita Family	DOSM PriceCatcher
Produce	Fruits & Vegetables	Produce, Vegetables, Fruits	`PRODUCE`	Sayur-Sayuran, Buah-Buahan
Dairy	Dairy	Dairy & Eggs	`DAIRY`	Susu & Produk Tenusu
Meat	Meat & Poultry	Meat, Poultry, Seafood	`MEATS, POULTRY, SEAFOOD`	Daging Segar, Ayam, Laut
Bakery	Bakery	Bakery & Bread	`BREAD/BAKERY`	Roti & Barangan Bakeri
Pantry	Canned Goods, Grains	Dry Goods, Canned	`GROCERY I, GROCERY II`	Barangan Kering, Beras

4.2 Goal 4 Full — Discrete-Event Simulator Engine (`simulation_events.csv`):
Mensimulasikan dinamika surplus 3–5 supermarket di Lembah Klang selama horizon 30–90 hari. Sesuai arahan arsitektur retail (Step 7–14), **DILARANG MENGGUNAKAN EVENT PERTANIAN (Harvested/Shipped)**. Engine hanya menggunakan 5 siklus event ritel resmi:

```
[ received ] → [ active_fefo ] → [ markdown ] → [ dispatched ] (Food Bank via OSRM)
■ [ wasted ] (Landfill / Rusak)
```

5. Cross-Functional Team Interoperability Contract Matrix

Person 3 berinteraksi langsung dengan seluruh rekan tim melalui kontrak data formal berikut:

Team Member	Flow	Artifact / Endpoint	Contract Schema & Fields	Integration Guarantee
Person 0 (Dataset Quality)	**IN**	`Grocery_Inventory_cleaned, holidays_my.csv, item.parquet`	Typo Catagory diperbaiki, Cabbage diisi 'Produce', tabel lookup DOSM MYR.	Person 3 mengonsumsi file bersih sebagai basis ETL.
Person 1 (Tech Lead & ML)	**OUT**	`data/processed/lots_base.csv`	shelf_life_days, turnover_ratio, stock_pressure, category, days_to_expiry.	Format 100% kompatibel dengan src/ml/spoilage.py.
Person 2 (Forecaster & API)	**OUT**	`lots_base.csv` → `GET /api/lots`	lot_id, product_name, category, quantity, expiry_date, store_id.	Person 2 membaca CSV untuk melayani endpoint REST API.
Person 4 (Logic & Supabase)	**OUT**	`Supabase Table: batches (via upload_batches.py)`	Seluruh kolom lots_base.csv di-upsert ke tabel PostgreSQL dengan RLS.	Zero schema mismatch; kolom Upgrade disiapkan NULL.
Person 5 (Risk & OSRM)	**OUT**	`lots_base.csv` → `OSRM Dispatch Engine`	lot_id, quantity (kg), category, dan lokasi toko untuk rute OSRM.	Menjamin kuantitas surplus valid untuk kapasitas food bank.
Person 6 (Frontend UI)	**OUT**	`Tabel Stok FEFO & Web App React/Vite`	Format DTE, label risiko, nama produk, dan harga satuan dalam Ringgit Malaysia (RM).	Langsung selaras dengan inventoryData.js frontend.

6. Implementation Roadmap, Deliverables Checklist & Sign-Off Matrix

No	Task Deliverable Description	Phase	Target File / Artifact	Status
1	Pembersihan currency $, parsing tanggal ISO, dan kalibrasi anchor date.	**BASE**	`src/etl/process_store_inventory.py`	**VERIFIED**
2	Ekspor dataset stok retail bersih 1.000 lot siap konsumsi Supabase.	**BASE**	`data/processed/lots_base.csv`	**READY**
3	Script validasi integritas otomatis (7 Golden Rules test suite).	**BASE**	`src/etl/validate_lots_base.py`	**READY**
4	Harmonisasi skema kanonikal 5 dataset pangan heterogen.	**UPGRADE**	`canonical_batch_table.csv`	PLANNED
5	Multi-Store Discrete-Event Simulator Engine (5 event ritel).	**UPGRADE**	`simulation_events.csv`	PLANNED

PERNYATAAN RESMI KONFIRMASI TEKNIS (FORMAL SIGN-OFF & AUTHORIZATION):
Dengan diterbitkannya dokumen konfirmasi teknis ini, **Person 3 (Data Architect & Pipeline Lead)** menyatakan kesiapan 100% untuk mengeksekusi pipeline data sesuai spesifikasi di atas. Arsitektur data telah diverifikasi selaras dengan model ML Person 1, kebutuhan database Supabase Person 4, OSRM Dispatch Person 5, dan web dashboard Person 6. Eksekusi Fase 1 (BASE) dimulai secara mandiri tanpa bottleneck.